AI-Driven Projection Mapping for Ambient Spatial Computing

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github.com/namanrajpal/LightMapAI

Abstract

We present LightMapAI, an open-source system that combines AI-driven shader generation with real-time projection mapping to enable ambient spatial computing. Users describe visual intent in natural language; the system synthesizes GPU shaders and projects them onto calibrated physical surfaces. Our architecture introduces a proxy-node animation model that bridges Godot Engine's AnimationPlayer to runtime surface property control, enabling live keyframe animation of opacity, color, geometry, and shader parameters without editor dependency. The system performs real-time perspective homography warping with pixel-precise calibration across multiple surfaces.

Motivation

Commercial projection mapping tools (MadMapper, Resolume) require expensive licenses and steep learning curves. Artists and researchers with a projector and a creative vision shouldn't need VJ expertise to transform physical spaces. LightMapAI makes ambient spatial computing as simple as describing what you want to see.

System Architecture

- **SurfaceManager** — Central data store for all projection surfaces (position, color, opacity, corners)
- **SurfaceProxy** — Bridge nodes with exported properties that AnimationPlayer targets; setters forward changes to SurfaceManager
- **TimelineManager** — Autoload hosting AnimationPlayer, proxy lifecycle, keyframe CRUD, playback control, JSON serialization
- **AI Shader Generation** — Natural language prompts generate GLSL fragment shaders for surface content via LLM integration
- **Perspective Homography** — Real-time 3×3 matrix inversion maps screen coordinates back to UV space for accurate warping
- **Dual Output** — Setup UI on laptop, clean projection on secondary display simultaneously

Demo

Live demonstration projecting onto a physical lamp with three independently mapped surfaces. Each surface displays a unique AI-generated shader effect created through natural language prompts. Attendees can interact hands-on: create surfaces, warp them onto objects, generate visuals via AI, and animate properties in real time.

Key Contributions

- AI shader generation from natural language — describe a visual, project it onto physical space
- Proxy-node animation architecture enabling runtime keyframe editing without Godot's editor
- Open-source alternative to commercial projection mapping software (under 3MB)
- Pixel-precise corner calibration with real-time homography computation
- JSON serialization for portable, shareable show files
- Animated opacity, color, visibility, z-order, and geometry with 5 easing types

Future Work

- OSC/MIDI integration for live performance control
- Audio-reactive shader parameters driven by FFT analysis
- Bezier edge warping for curved surface mapping
- Multi-projector edge blending
- Cue list sequencer for show automation
- BPM sync for music-driven performances

Open Source

Built with Godot Engine 4.x

GDScript · GL Compatibility · Cross-platform

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